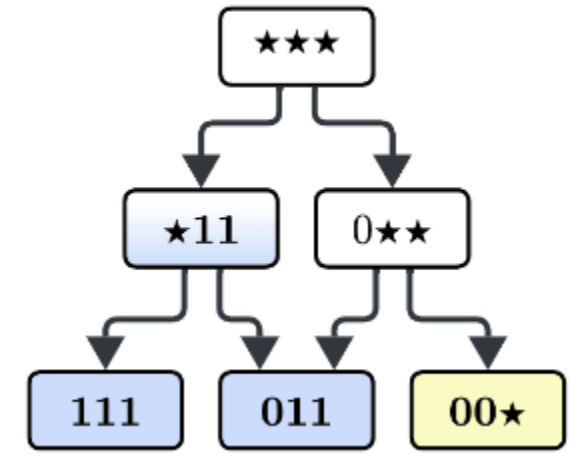
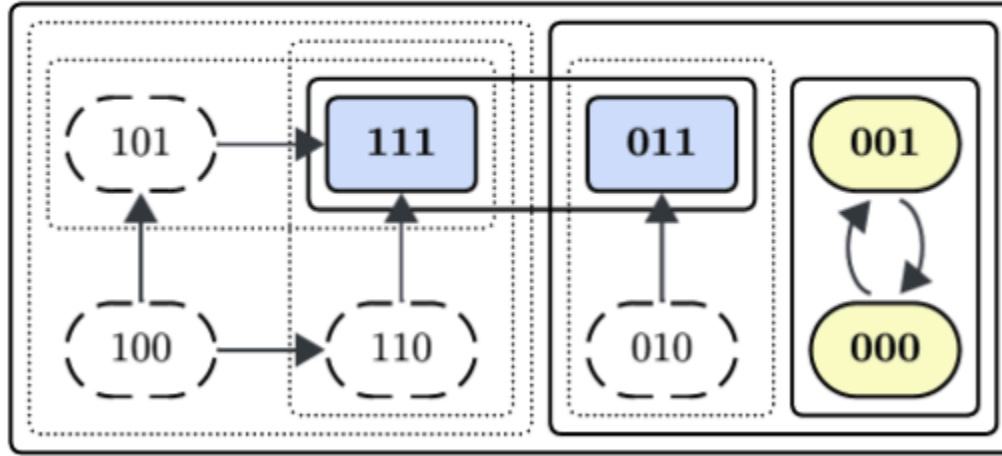
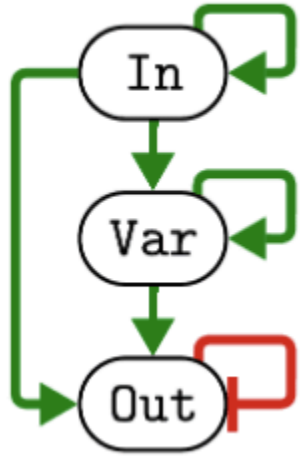


**Relating biomarkers and
phenotypes using
dynamical trap spaces**

Boolean models!



The influence graph can be anything; but in practice, it often has this specific (input $\Rightarrow$ model $\Rightarrow$ output) hierarchy, possibly with many inputs and outputs.

What is a phenotype?

(not again...)

- A useful Boolean model should explain some "high-level behavior" of the biological system.
- Is **attractor** a phenotype? Maybe, but we can have 1.000s or 1.000.000s of attractors (esp. with many inputs), and only a few actual phenotypes. Same goes for **minimal trap spaces**.
- We rely on **phenotype determining nodes** (PDNs; biomarkers).
 - Different values of PDNs = different phenotype.

What is a phenotype?

(not again...)

- How do we select PDNs for our model? How do we validate that our selection is meaningful?
 - Just take the output nodes, right? (maybe?)
 - Literature search all the things!
 - Can we do better?
- What can we do with PDNs once we have them?
 - What is "the feature" of a Boolean model that we are actually looking for?

Selecting PDNs

- We can select PDNs manually $\Rightarrow$ We should know what the model was supposed to tell us.
- But are our assumptions correct? Did we actually model what we expected to model?
- What if we could compare our manual selection with some "objectively good" PDN set?

Selecting PDNs (mutual information)

- Assume we have the knowledge of all attractors.
- Any PDN set partitions the attractors into (up to) 3^k groups.
- We can compute the **mutual information** between the PDN node values and the values of remaining nodes across these groups.
- Intuitively: *If I only observe the PDN values, can I predict the state of the rest of the network?*
- Algorithmically, testing all possible PDN sets is impractical, so we tested a simplified greedy approach $\Rightarrow$ Pick one PDN at a time, always taking the greedy choice that maximizes information gain.

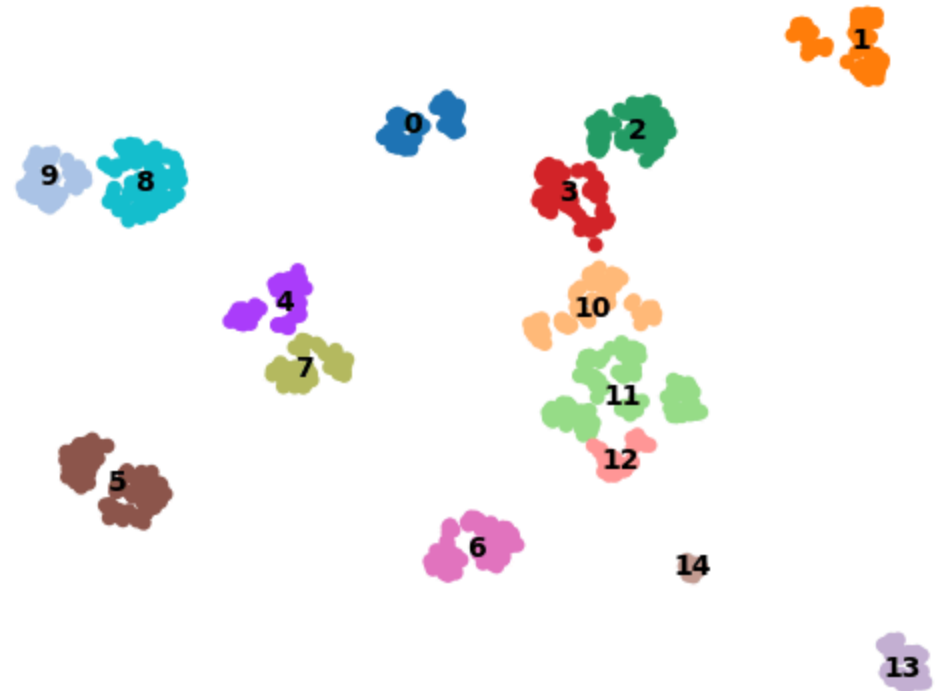
Selecting PDNs (attractor clustering)

- Assume we have the knowledge of all attractors.
- Assume each attractor is a "single cell" in a simulated RNA-like dataset.
- How would I select significant biomarkers based on such a dataset?
 - Dimensionality reduction (PCA $\Rightarrow$ MCA)
 - Unsupervised clustering (Leiden algorithm)
 - Feature importance selection
- Intuitively: *If I only observe the PDN values, can I predict which cluster an attractor belongs to?*
- Needs sensitivity testing; unsupervised clustering is inherently probabilistic.

Selecting PDNs (attractor clustering)

MCA also provides a flawed but viable way to visualize the attractor clusters.

UMAP of MCA Coordinates and Leiden clusters

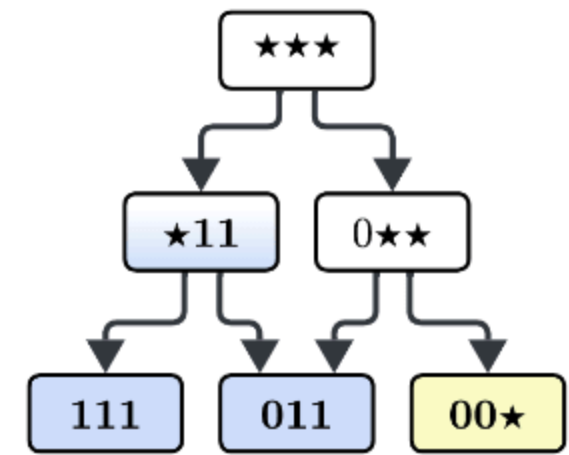
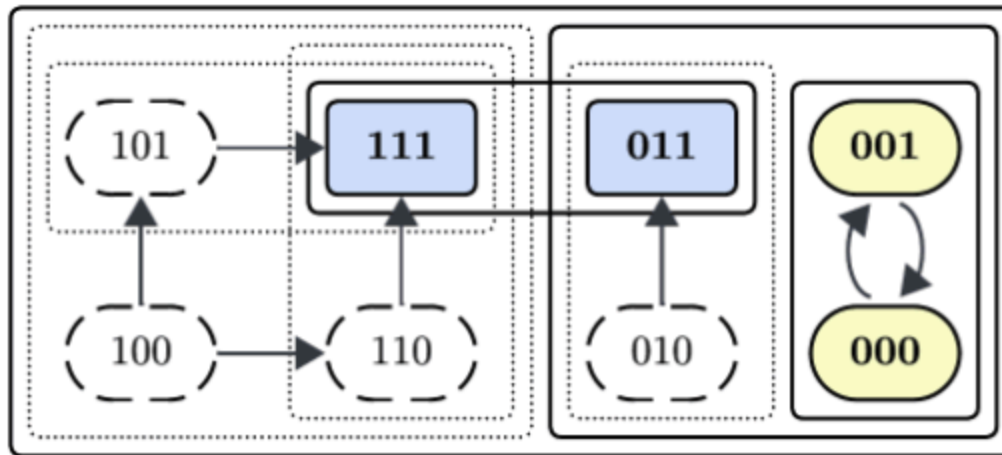
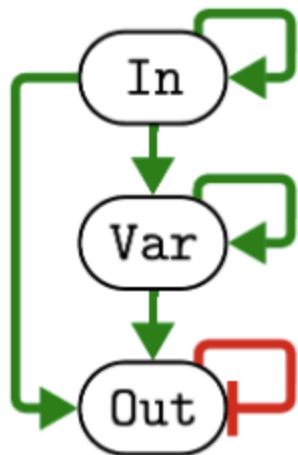


Selecting PDNs (LDOI)

- So far, we required the knowledge of all attractors (or at least some representative sample).
- Logical domain of influence (LDOI) of a subspace S :
 - Let's fix model variables according to S .
 - Repeatedly propagate all values that are universally fixed across the whole subspace (percolation).
 - The resulting subspace (or rather the newly fixed variables) is the LDOI.
- LDOI PDNs: Select PDNs such that across the 3^k possible subspaces, we maximize the average LDOI.
- Intuitively: Select nodes that when fixed, cause the rest of the network to become fixed as well.
- A non-local measure of "canalyzation" in the model. We want to know how much a node influences the rest of the network, not just its direct regulation targets.

So what is a phenotype?

- Once we know the PDNs, how do we use them to analyze a model?
- Dynamical phenotype subspace is:
 - A trap space (cannot be escaped)
 - Fully percolated (all values are fully propagated)
 - No smaller subspace in it has a different combination of PDN values.
- Every such subspace represents a fully stabilized "phenotype", and describes the minimal initial conditions to achieve that phenotype.



Case Study

- A T cell differentiation model (70 nodes; known literature-based PDNs).
- Our three heuristics mostly agree on what is important.
- They don't agree that much with our literature-based PDNs (TBET, GATA3, FOXP3, RORGT, NFAT, IFNG, IL4, IL17, and TGFB).
- Notably, the literature-based PDN set is much larger!

Rank	LDOI	clustering-based	mutual information
1	IL4R_b1	IL4R_b1, STAT6, IL12RB2	TBET, RUNX
2	NFAT	STAT1, IRF1, IL12RB1	NFAT, IL2RA, IKB
3	TBET	NFAT, IL2RA, IKB	SMAD3, TGFBR
4	STAT1	TBET, RUNX	IL4R_b1, STAT6, IL12RB2
5	proliferation	SMAD3, TGFBR	IL4RA, STAT5_b2
6		IL4RA, STAT5_b2	STAT1, IRF1, IL12RB1
7		IL12R, STAT4	IL23R
8		FOXP3, NFKB	FOXP3, NFKB

Case Study

Bottom line: The original study "zooms in" on a specific sub-category of attractors; Meanwhile, we are getting the "optimum" over the whole attractor repertoire.

	resting, not Th1				resting, Th1				active/anergic, Th1				active/anergic, not Th1			
	not proliferating		proliferating		not proliferating		proliferating		not proliferating		proliferating		not proliferating		proliferating	
	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
Th2_re	1	1	1	1	1	2	1	1								
Th2-Th17_re	1	2	2	1	1	4	2	1								
Th0_re				1				1								
Th17_re				1				1								
Th1_re									1	1	1	1				
Th1-Th17_re									1	2	2	1				
Th1-Treg_an									1	1	1	1				
Th1-Treg-Th17_an									1	1	1	1				
Th1_an									1	1	1	1				
Th1-Th17_an									1	3	2	1				
Th1-Treg_ac_TGFB													1	1	1	1
Th1-Treg-Th17_ac_TGFB													1	3	2	1
Th1_ac													1	1		
Th1-Th17_ac_IFNG													1	1		
Th1-Th17_ac_IL17															1	
Th2_an													1	1		
Th2-Th17_an													1	1		
Th2-Treg_an													1	1	1	1
Th2-Treg-Th17_an													1	1	1	1
Treg_an															1	
Treg-Th17_an															1	
Th2_ac															1	
Th2-Th17_ac_IL4															1	
Th2-Treg-Th17_ac_TGFB															1	
Th0_ac																1
Th17_an																1
Treg_ac																1
Treg-Th17_ac_TGFB																1
Treg_ac_o																1
Treg-Th17_ac_TGFB_o																1

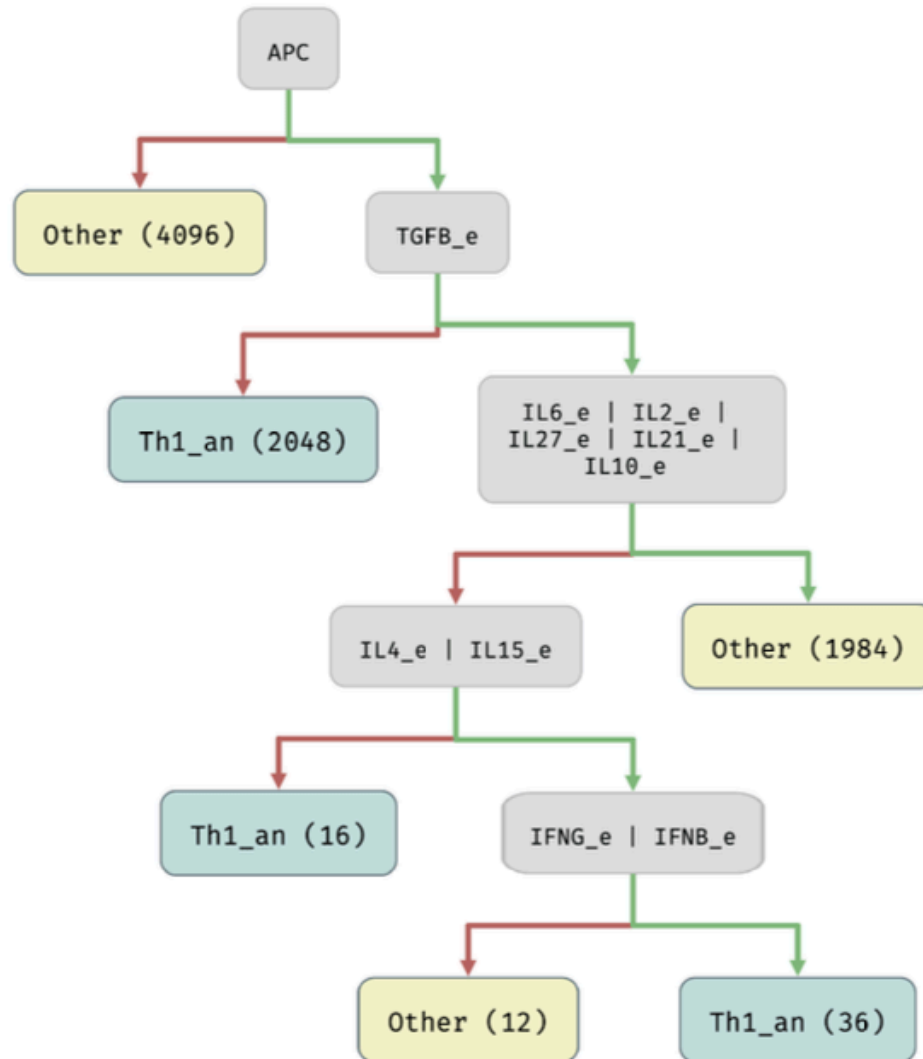
NFAT		TBET		
0	resting	0	not Th1	
0	resting	1	Th1	
1	active/anergic	0	not Th1	
1	active/anergic	1	Th1	
IL4R_b1		STAT1		
1		0		A
1		1		B
0		1		C
0		0		D

Case Study

LDOI \ cluster	25	20	15	16	4	5	6	12	13	14	7	9	8	10	30	0	1	11	17	2	3	18	26	19	29	23	27	21	22	28	24
11010	3																														
11110		3																													
00010			2	1																											
00011			2	1																											
10110					2	1																									
10111					2	1	3																								
10010								2	1																						
10011								2	1	3																					
00110											1	1	1	1																	
00111											1	1	1	1																	
01010															1																
10100																1															
10101																1	1														
10001																	1	1													
10000																		1													
11000																			1												
11100																			1												
00000																				1	1										
00001																				1	1										
00100																				1	1										
00101																				1	1										
01000																					1										
01100																					1	1									
01001																					1										
11001																						2									
01011																							3								
11011																								3	2						
01101																									3	5					
11101																									1	1	1				
11111																									3	5	1				1
01111																									2	2	1				1
01110																											1	1	1	1	

Meanwhile, the automated selection methods largely agree with each other.

Case Study



If we know the phenotype subspaces, we can describe the inputs and initial conditions that enable (or guarantee) the phenotype.

In the case study, almost all phenotypes have trivial input conditions. For the rest, we can use a decision tree.

Computing dynamical phenotypes

- No time :(Let's discuss after the talk!

Takeaways

- We can try to "infer" PDNs from network attractors; but we need to know the attractors.
- We can use LDOIs to select "most influential" PDN nodes in a network (without relying on all attractors).
- Does not necessarily mean inferred PDNs are "better"... they capture a specific notion of "phenotype", but may need additional tuning.
- In our case, we can use much fewer variables to capture roughly the same amount of "variability" in the data.
- Possible validation that your notion of phenotype agrees with what is in fact significant in the model.